Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 5 · SLO 2 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 5 : SLO â€“ 2 | Singly Linked List â€” Insertion at the Beginning and End

I. Insert a New Node VAL at the Beginning

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 7
- [END OF IF]
- Step 2: SET NEW_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW_NODE -> DATA = VAL
- Step 5: SET NEW_NODE -> NEXT = START
- Step 6: SET START = NEW_NODE
- Step 7: EXIT

II. Insert a New Node at the End

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 10
- [END OF IF]
- Step 2: SET NEW_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW_NODE -> DATA = VAL
- Step 5: SET NEW_NODE -> NEXT = NULL
- Step 6: SET PTR = START
- Step 7: Repeat Step 8 while PTR -> NEXT != NULL
- Step 8: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 9: SET PTR -> NEXT = NEW_NODE
- Step 10: EXIT